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Random Attention: Rethinking KV Cache Eviction for Efficient Reasoning

Scoring Cached Tokens Adds Nothing — Random Eviction with Prompt Protection Matches the Best Evictors at 32–43% Higher Throughput

By Heng Wang, Jieliu Qiu, Wenting Zhao, Cheng Qian, Liangwei Yang, Jiawei Han, Heng Ji, Silvio Savarese, Shelby Heinicke, Huan Wang (Salesforce AI Research; UIUC; Cornell; et al.)

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Long chain-of-thought reasoning turns the KV cache into a severe memory bottleneck. Every token in a reasoning trace adds a key-value pair; at thousands of tokens per query, the cache overflows batch capacity and degrades throughput. Every existing KV cache compression method responds with the same paradigm: score each cached token by some proxy for future importance, then keep the top scorers. Random Attention discards the scores entirely. It keeps the prompt tokens and evicts all others uniformly at random — and across four models and six reasoning tasks matches the best-performing scored evictors while achieving 32–43% higher throughput in vLLM.

AT A GLANCE

Problem: Long CoT reasoning makes the KV cache a memory bottleneck; existing eviction methods rely on expensive importance scoring.

Why it matters: Production inference systems serving reasoning models need maximal throughput; scoring overhead is non-trivial.

Method: Keep prompt KV cache; evict reasoning-trace tokens uniformly at random per attention head. No score computed.

Result: Matches strongest evictors on 6 tasks; 32–43% higher throughput at 50% cache budget in vLLM.

Evidence: Ablations isolating prompt protection from scoring; redundancy analysis across text and attention heads.

The Central Observation

Every KV cache evictor in the literature shares a core design assumption: the identity of which tokens to retain matters, and a good scoring function is needed to identify them. H2O uses cumulative attention weight; StreamingLLM uses recency with attention sinks; LESS

uses gradient-based estimates; SnapKV clusters similar keys.

Random Attention shows this assumption is false, at least for long reasoning traces. The scoring signal contributes almost nothing to the final task performance, once you control for whether the prompt is preserved.

Why the Scoring Signal Is Worthless

The authors run controlled ablation experiments across all four model families:

Condition A: Prior evictor, prompt not explicitly protected — the prompt may happen to be evicted.

Condition B: Prior evictor with explicit prompt protection.

Condition C: Random eviction, no prompt protection.

Condition D (Random Attention): Random eviction with prompt protection.

The results show Condition B $\approx$ Condition D $\gg$ Conditions A and C. The gap between A and D is almost entirely explained by whether the prompt is preserved, not by whether scoring is used. The gap between B and D (evictor scoring vs. random after prompt protection) is negligible.

Two Mechanisms Explaining Redundancy

Why does random eviction of the reasoning trace lose almost nothing?

Text-level redundancy. Models engaged in chain-of-thought reasoning restate the facts they need as they work. If the model needs P to derive Q three steps later, it typically restates P (or a consequence of P) at the derivation site. Evicting the original occurrence of P loses nothing because the restatement survives. The authors measure text-level redundancy: roughly 60% of trace tokens can be deleted while preserving QA-measured semantic content.

Head-level redundancy. Each attention head maintains its own copy of the KV cache. Information encoded at a given token is distributed across heads. When eviction is performed independently per head (as in Random Attention), the probability that a fact encoded in m heads is retained by at least one head is:

$$P(\text{fact retained}) = 1 - (1 - p)^m$$

For $m = 6$ heads and $p = 0.5$ budget: $P = 1 - 0.5^6 = 0.984$. This near-certainty of retention means no scored selection is needed.

Technical Formulation

Let $\mathcal{P}$ be the set of prompt token positions and $\mathcal{T}$ the set of reasoning-trace positions. Given a cache budget B ,

standard evictors compute:

$$\mathcal{R} = \mathcal{P} \cup \text{top}_k^{\text{score}}(\mathcal{T}), \quad k = B - |\mathcal{P}|$$

Random Attention replaces scored selection with uniform sampling:

$$\mathcal{R}^{(h)} = \mathcal{P} \cup \text{sample}_k^{\text{uniform}}(\mathcal{T})$$

performed independently per attention head h . No score function is defined or computed.

Experimental Findings

Experiments span four models — DeepSeek-R1-Distill-Qwen-7B, DeepSeek-R1-Distill-Qwen-14B, QwQ-32B, and DeepSeek-R1-32B — and six tasks: AIME 2024, AIME 2025, AMC, MATH500, LiveCodeBench, and GPQA Diamond. At 50% cache budget:

Method	AIME 2024	MATH500
Full KV cache	72.3%	91.4%
H2O (50%)	68.1%	89.3%
StreamingLLM (50%)	67.4%	88.7%
Random Att. (50%)	68.4%	89.5%

Random Attention matches or slightly exceeds scored evictors, while serving 32–43% higher throughput in vLLM deployment.

Throughput Analysis

The throughput advantage has two sources. First, eliminating the score computation removes a matrix product over the full KV sequence per generation step, saving roughly 15% of the per-step compute at 50% budget. Second, uniform random eviction produces better memory access patterns than score-sorted eviction (which leaves a non-contiguous, score-ranked subset in the cache), saving the remaining 17–28%.

Ablations and Interpretation

Per-head vs. global eviction: Per-head independent sampling outperforms global uniform eviction by 1.2 AIME points, confirming that head-level redundancy requires per-head independent treatment.

Tight budgets (25%): Random Attention is more robust than scored evictors at 25% budget, where its MATH500 accuracy matches scored evictors at 50%.

Task type: Temporal and procedural tasks (LiveCodeBench) benefit slightly more from Random Attention than purely mathematical tasks, consistent with higher restatement frequency in procedural traces.

Prompt fragility confirmed: If the prompt is evicted even partially (Condition C), accuracy drops 17+ points

on AIME 2024 — confirming that the prompt, not the trace, is semantically irreplaceable.

Reference

Heng Wang et al. **Random Attention: Rethinking KV Cache Eviction for Efficient Reasoning.** arXiv:2609.03430, EMNLP 2026. <https://arxiv.org/abs/2609.03430>